PGCP-AI • AUGUST 2026 • EXAM PREPARATION

DATA ANALYTICS

Complete Notes and Lab Revision Guide

Built around the ACTS, Pune Suggested Teaching Guidelines for Data Analytics

Course facts from the supplied syllabus

Duration: 120 hours = 50 theory + 50 lab + 20 self-learning. Evaluation: Theory exam 40%, Lab exam 40%, Internal assessment 20%. Prerequisites: Python programming and basic statistics.

Designed for fast revision: connected explanations, formulas, worked examples, Python lab patterns, exam traps, interview-style questions, and a self-test with answers.

Study principle: understand the question -> inspect the data -> choose the method -> check assumptions -> interpret the result in business language.

How to use these notes

These notes expand the topics named in the supplied 4-page syllabus. The organizing sequence follows the syllabus, while examples, formulas, Python patterns, and revision questions are added to make the topics easier to study. Where the syllabus only names a topic (for example, directional or functional data analysis), the explanation is intentionally introductory.

Recommended revision loop

- Pass 1 - read the connected concept flow and the comparison tables.
- Pass 2 - reproduce every formula and worked example without looking.
- Pass 3 - run the lab snippets on a small dataset and explain each output.
- Pass 4 - attempt the final quiz; revisit every wrong answer and its assumption.

Course map

Syllabus block	Topics covered in these notes
Sessions 1-3	Business analytics, analytics lifecycle, intelligent data analysis, data types, tools, reporting vs analysis
Sessions 4-5	EDA, visualization, central tendency, dispersion, quartiles, percentiles, CV
Sessions 6-9	Samples/populations, probability, Bayes, distributions, CLT, sampling, estimation
Sessions 10-12	Covariance, correlation, outliers, hypothesis testing, Z-test, chi-square, skewness
Sessions 13-16	Predictive modelling, informative attributes, supervised segmentation, trees, probability estimation, prescriptive modelling
Sessions 17-20	Regression, forecasting, simulation, risk analysis, linear and nonlinear optimization
Sessions 21-23	Overfitting, generalization, holdout vs cross-validation, classifiers, decision analytics, evidence and Bayes reasoning
Session 24	Factor analysis, directional data analytics, functional data analysis
Session 25	Power BI architecture, Power Query, visualizations, reports, dashboards, sharing insights

The full analytics chain

Remember the order

Business question -> define target and success measure -> collect/understand data -> clean and prepare -> explore and visualize -> plan/select model -> train -> validate -> communicate -> approve/deploy -> monitor and operate.

01. Business analytics and the analytics lifecycle

Business analytics converts data into decisions. A useful analysis starts from a real decision, not from a chart or algorithm.

1.1 Analytics types

Type	Question answered	Example
Descriptive	What happened?	Revenue fell 8% last month.
Diagnostic	Why might it have happened?	Revenue fell because visits dropped in one region.
Predictive	What is likely to happen?	Demand is forecast to rise next week.
Prescriptive	What should we do?	Shift inventory to regions with forecast shortages.

Memory trick

Describe -> diagnose -> predict -> prescribe. These differ by the question, not by whether a chart or a model is used.

1.2 Analytics lifecycle in the syllabus

- Discovery: clarify business context, stakeholders, constraints, available data, and success criteria.
- Data preparation: collect, join, clean, validate, transform, and document data.
- Model planning: choose an analytic approach, target, features, validation plan, and evaluation metric.
- Model building/implementation: develop the analysis or model, tune it, and record parameters.
- Quality assurance: test data quality, code, assumptions, reproducibility, and model performance.
- Documentation and management approval: explain methods, limitations, risks, and business consequences.
- Installation/deployment: place the approved solution in its intended environment.
- Acceptance and operation: verify it works for users, then monitor data and performance over time.

1.3 Analysis vs reporting

Reporting	Analysis
Usually answers what happened.	Investigates why, what may happen, or what to do.
Often recurring and standardized.	Often exploratory and question-driven.
Example: weekly sales dashboard.	Example: test whether a price change affected sales after controlling for season.

1.4 Quality gates

- Data: correct types, valid ranges, missingness understood, duplicate policy defined.
- Method: assumptions and metric match the question; no target leakage.
- Result: baseline comparison, uncertainty, and limits disclosed.
- Delivery: reproducible code, versioned data/model, documentation, access controls, and monitoring plan.

Lab setup

PYTHON / PSEUDOCODE

```
# In Google Colab or Jupyter
import numpy as np
import pandas as pd
import matplotlib.pyplot as plt
import seaborn as sns
from scipy import stats
from sklearn.model_selection import train_test_split, cross_val_score
from sklearn.pipeline import Pipeline
from sklearn.preprocessing import StandardScaler
from sklearn.linear_model import LogisticRegression
```

02. Intelligent data analysis, data nature, and EDA

EDA (exploratory data analysis) is the first disciplined conversation with a dataset. It helps reveal structure, errors, unusual values, and hypotheses worth testing.

2.1 Types of data

Type	Meaning	Example / useful summary
Numerical - discrete	Countable values	Number of orders; counts, bar plot
Numerical - continuous	Measurements on a scale	Delivery time; histogram, mean/SD
Categorical - nominal	Labels without natural order	City, product category; counts/proportions
Categorical - ordinal	Labels with order	Low/Medium/High; ordered frequencies
Binary	Two possible classes	Churn yes/no; proportion and class balance
Time series	Values indexed by time	Daily sales; trend/seasonality
Spatial/geographic	Location-dependent data	Coordinates, region; maps / spatial patterns

2.2 Shape of a dataset

PYTHON / PSEUDOCODE

```
df.shape                # rows, columns
df.head()               # first records
df.info()               # types and non-null counts
df.describe(include="all")
df.dtypes
df.isna().sum()         # missing values per column
df.duplicated().sum()   # duplicated rows
df.nunique()            # cardinality per column
```

2.3 EDA workflow

- Check dimensions, columns, data types, units, and sample records.
- Check missing values, duplicates, impossible values, and inconsistent labels.
- Summarize numeric columns; count categories; inspect class imbalance.
- Visualize distributions, relationships, time patterns, and outliers.
- Write an observation, a plausible explanation, and what evidence would verify it.

2.4 Chart selection

Question	Good first choice	Watch out for
Distribution of one numeric variable	Histogram, density plot, box plot	Bins and extreme values can change appearance.
Compare numeric value by group	Box/violin plot, mean with interval	Groups may have different sample sizes.
Compare categories	Bar chart	A bar chart is for category counts/aggregates, not a continuous distribution.
Relationship between two numeric fields	Scatter plot	Correlation can be driven by outliers or a third variable.
Trend over time	Line chart	Sort time correctly; watch seasonality and missing periods.
Association between categories	Stacked/clustered bars, heatmap	Use proportions when category totals differ.

2.5 Hypotheses from visuals

Example: a box plot suggests mobile users spend less per order than desktop users. This is an observation, not proof. A possible test is H0: the relevant population means/distributions are equal; H1: they differ. Check confounders such as region, user type, device mix, and campaign source before claiming causality.

Exam trap

A chart can suggest a pattern, but it does not by itself establish causation or statistical significance.

03. Descriptive statistics and summary measures

Descriptive statistics compress a dataset into interpretable numbers. Always pair a location measure with a measure of spread, and inspect the shape.

3.1 Central tendency

Measure	Definition / formula	Best use and limitation
Mean	sum(x) / n	Uses every value; sensitive to extreme outliers.
Median	Middle sorted value; average of two middles if n is even	Robust to extreme values; useful for skewed income/prices.
Mode	Most frequent value/category	Useful for categorical data; can have multiple modes.

3.2 Dispersion

Measure	Formula / meaning	Interpretation
Range	max - min	Very sensitive to extremes.
Quartiles	Q1 = 25th percentile; Q2 = median; Q3 = 75th percentile	Show the distribution in four parts.
IQR	Q3 - Q1	Spread of the middle 50%; robust.
Population variance	sum((x - mu)^2) / N	Average squared deviation from population mean.
Sample variance	sum((x - xbar)^2) / (n - 1)	Uses Bessel correction for estimating population variance.
Standard deviation	Square root of variance	Spread in the same unit as the data.
Coefficient of variation	SD / mean x 100%	Relative spread; only sensible for a meaningful nonzero mean on a ratio scale.

3.3 Worked example

Data: 2, 4, 4, 6, 9. Mean = (2+4+4+6+9)/5 = 5. Median = 4. Mode = 4. Range = 9-2 = 7. Using the common median-of-halves convention, Q1 = 3 and Q3 = 7.5, so IQR = 4.5. Quartile conventions can vary by software; use the convention specified in the question or library.

3.4 Percentiles and quartiles

- The p-th percentile is a value at or below which roughly p% of observations lie, depending on the chosen quantile definition.
- Q1 = P25, Q2 = P50, Q3 = P75. A percentile rank answers “what percentage is at/below this value?”
- Different software may interpolate percentiles differently, especially for small samples. State the method if precision matters.

3.5 Outlier fences

PYTHON / PSEUDOCODE

```
IQR = Q3 - Q1
lower_fence = Q1 - 1.5 * IQR
upper_fence = Q3 + 1.5 * IQR
# Points outside the fences are flagged, not automatically errors.
```

3.6 Welford online variance - exam idea

Welford’s algorithm updates mean and the sum of squared deviations one observation at a time. It is numerically more stable than repeatedly computing sum(x^2) - n*mean^2 and is useful for streaming data.

PYTHON / PSEUDOCODE

```
n = 0
mean = 0.0
M2 = 0.0
for x in values:
    n += 1
    delta = x - mean
    mean += delta / n
    delta2 = x - mean
    M2 += delta * delta2
sample_variance = M2 / (n - 1) if n > 1 else float("nan")
```

Memory trick

Mean gets pulled by extremes; median resists extremes. Variance is squared units; standard deviation returns to original units. IQR describes the middle 50%.

04. Probability, samples, events, and Bayes theorem

Probability provides a language for uncertainty. Sample spaces describe possible outcomes; events are subsets of those outcomes.

4.1 Population, sample, and sampling units

- Population: all units relevant to the question. Sample: observed subset used to learn about the population.
- A parameter describes a population (mu, sigma, population proportion p). A statistic describes a sample (x-bar, s, sample proportion p-hat).
- Univariate sampling concerns one variable at a time; bivariate sampling/analysis studies two variables jointly. Resampling repeatedly draws from observed data or generates alternative samples to estimate variability or evaluate a method.

4.2 Basic rules

Rule	Formula
Complement	P(not A) = 1 - P(A)
Addition	P(A union B) = P(A) + P(B) - P(A intersection B)
Conditional	P(A|B) = P(A intersection B) / P(B), when P(B)>0
Independence	A and B independent iff P(A intersection B)=P(A)P(B)
Joint probability	P(A and B)
Marginal probability	P(A), obtained without conditioning on another variable; often by summing joint probabilities

4.3 Bayes theorem

Formula

P(A|B) = P(B|A) P(A) / P(B). Posterior = likelihood x prior / evidence.

Example: a condition affects 1% of people. A test has 90% sensitivity and 5% false-positive rate. For 10,000 people, approximately 100 have the condition; 90 test positive. Of 9,900 without it, 495 test positive. Positive tests total 585, so P(condition | positive) = 90/585 = 15.4%, not 90%. This is the base-rate effect.

4.4 Naive Bayes

Naive Bayes uses Bayes theorem with a conditional-independence assumption among features given the class. This assumption is often unrealistic, but the classifier can still work well for text and high-dimensional sparse data.

PYTHON / PSEUDOCODE

```
from sklearn.naive_bayes import GaussianNB
from sklearn.model_selection import train_test_split
from sklearn.metrics import accuracy_score
X_train, X_test, y_train, y_test = train_test_split(
    X, y, test_size=0.2, random_state=42, stratify=y)
model = GaussianNB()
model.fit(X_train, y_train)
pred = model.predict(X_test)
print(accuracy_score(y_test, pred))
```

Exam traps

P(A|B) is usually not equal to P(B|A). Independence is different from mutually exclusive events. A low base rate can make positive predictive value low even with a seemingly accurate test.

05. Random variables, distributions, and the CLT

A random variable maps outcomes to numbers. A distribution describes how probability is spread across its possible values.

5.1 Discrete vs continuous

Concept	Discrete	Continuous
Possible values	Countable values, e.g. 0,1,2,...	Intervals of real values, e.g. time, height
Probability	P(X=x) may be positive	For an ideal continuous variable, P(X=exactly x)=0
Function	PMF (probability mass function)	PDF (probability density function)
Total probability	Sum of probabilities = 1	Area under density = 1

5.2 Key distributions

Distribution	Use / parameters	Mean and variance
Binomial	Number of successes in n independent trials with fixed success probability p	Mean np; variance np(1-p)
Poisson	Count of events in a fixed interval at rate lambda, under suitable assumptions	Mean lambda; variance lambda
Normal	Symmetric continuous bell curve, parameters mu and sigma	Mean mu; variance sigma squared

PYTHON / PSEUDOCODE

```
from scipy import stats
import numpy as np
# Binomial: probability of exactly 3 successes in 10 trials, p=0.2
stats.binom.pmf(3, n=10, p=0.2)
# Poisson: P(X <= 2) when expected count lambda=1.5
stats.poisson.cdf(2, mu=1.5)
# Normal: probability below z=1
stats.norm.cdf(1)
```

5.3 Normal distribution and z-score

A z-score tells how many standard deviations a value lies from the mean: z = (x - mu) / sigma. For a sample-based standardized score, use x-bar and s as appropriate. Standardization alone does not make data normally distributed.

68-95-99.7 rule

For an approximately normal distribution: about 68% lies within 1 SD of the mean, 95% within 2 SD, and 99.7% within 3 SD.

5.4 Central Limit Theorem (CLT)

Under common conditions, the sampling distribution of the sample mean becomes approximately normal as sample size grows, even if the source population is not normal. It concerns the distribution of sample means, not a claim that the raw data become normal. Independence and finite variance are common conditions; required sample size depends on skewness and tails.

5.5 Quick distribution selection

- Yes/no outcome across a fixed number of independent trials -> Binomial.
- Number of events in a fixed exposure/time/area with stable rate -> Poisson model may fit.
- Continuous measurement with symmetric bell-shaped variation -> Normal may be a reasonable model.
- Always inspect whether model assumptions are reasonable; a named distribution is not automatically correct.

06. Sampling, estimation, and statistical interfaces

Sampling is how observations are selected; estimation is how sample evidence is used to infer unknown population quantities.

6.1 Sampling methods

Method	Idea	Main risk / use
Simple random	Every unit has a known equal selection chance.	Can be costly; requires a sampling frame.
Stratified	Sample within meaningful groups (strata).	Useful to ensure subgroup representation.
Cluster	Randomly choose groups, then sample units within them.	Often cheaper; units in a cluster may be similar.
Systematic	Select every k-th unit after a random start.	Risk if the list has a periodic pattern.
Convenience	Sample what is easy to reach.	Selection bias; may not represent population.
Bootstrap	Resample observations with replacement.	Estimates statistic variability; assumes observed sample is informative.

6.2 Sampling distribution and standard error

A sampling distribution is the distribution of a statistic over repeated samples. Standard error is the standard deviation of that sampling distribution.

Sample mean

If observations are independent with population SD sigma, SE(x-bar) = sigma / sqrt(n). When sigma is unknown, estimate with s / sqrt(n). Larger n generally reduces standard error at a square-root rate.

6.3 Point and interval estimates

- Point estimate: one best sample-based value, e.g. x-bar estimates population mean mu.
- Confidence interval: method that, over repeated samples, would cover the true parameter at the stated rate (for example, 95%). Do not interpret a frequentist 95% CI as a 95% probability that this fixed parameter lies in this already-computed interval.
- Margin of error = critical value x standard error, subject to the method and assumptions.
- A larger confidence level generally gives a wider interval; a larger sample generally gives a narrower interval.

6.4 What to check before generalizing

- Was the sample selected in a way that represents the target population?
- Are observations independent or clustered/repeated?
- Are missing responses likely to be systematically different?
- Is the sample size adequate for the required precision and subgroup questions?

07. Correlation, covariance, outliers, and skewness

Association summaries describe how variables move together. They do not by themselves prove causation.

7.1 Covariance and Pearson correlation

Measure	Meaning	Scale
Covariance	Positive if variables tend to deviate from their means in the same direction; negative if opposite.	Depends on the units of both variables.
Pearson r	Standardized linear association: covariance divided by SDs.	Between -1 and +1.

Pearson correlation near +1 indicates a strong positive linear relationship; near -1 indicates a strong negative linear relationship; near 0 indicates little linear relationship. It does not exclude a nonlinear relationship.

7.2 Important distinctions

- Correlation is symmetric: corr(X,Y) = corr(Y,X). Regression prediction is directional and depends on which variable is the target.
- Correlation can be affected strongly by outliers and restricted ranges.
- A lurking variable may drive both variables. Simpson’s paradox can occur when an aggregate relationship reverses after grouping by a confounder.
- Statistical association is not causal effect; design and causal assumptions matter.

7.3 Outlier handling workflow

- Detect: boxplot/IQR fences, z-scores, robust statistics, domain rules, or model residuals.
- Investigate: input error, valid rare event, process shift, data merge issue, or genuine heavy tail?
- Choose transparently: correct a confirmed error; keep a valid value; transform; cap/winsorize; use robust methods; or run a sensitivity analysis.
- Document rules and fit any learned thresholds on training data only when modelling.

Winsorization example: a high/low tail is capped at chosen percentile cutoffs. It reduces extreme influence but changes values; it is not the same as deleting rows. Use domain logic or justified cutoffs rather than blindly clipping unusual data.

7.4 Skewness

Shape	Interpretation	Common pattern
Approximately symmetric	Left and right sides are broadly balanced.	Mean and median may be close.
Right/positive skew	Long tail toward high values.	Mean often greater than median; income/spending often right-skewed.
Left/negative skew	Long tail toward low values.	Mean often less than median.

Exam trap

Do not remove an outlier solely because a rule flags it. First decide whether it is an error or a meaningful rare observation.

08. Hypothesis testing: Z-test and chi-square test

Hypothesis testing is a structured way to assess whether observed evidence is inconsistent with a stated null model, using a preselected significance level.

8.1 Test workflow

- State H0 (null) and H1 (alternative) in terms of the parameter or relationship.
- Choose alpha before looking at the result (often 0.05, but not universally).
- Check the design and assumptions; choose a suitable test statistic.
- Calculate the statistic and p-value, or compare to a critical value.
- If p <= alpha, reject H0. Otherwise, fail to reject H0; do not say H0 has been proved true.
- Interpret direction, effect size, confidence interval, practical significance, and limitations.

8.2 P-value and errors

Concept	Meaning
p-value	Assuming H0 and model assumptions hold, probability of results at least as extreme as observed under the test definition.
Type I error	Rejecting a true H0; long-run rate controlled by alpha under assumptions.
Type II error	Failing to reject a false H0; probability is beta for a specified alternative.
Power	1 - beta: chance of detecting a specified effect when it is present.
Practical significance	Whether the effect matters in the real decision; may differ from statistical significance.

8.3 Z-test (introductory pattern)

For a one-sample mean with known population SD sigma, Z = (x-bar - mu0) / (sigma / sqrt(n)). A Z procedure is also used for some proportions. In practice, if sigma is unknown and the population is estimated from a sample, a t-test is often the relevant mean test; follow the problem statement.

Worked structure

Suppose H0: mu = 100, n = 36, sample mean = 104, and known sigma = 12. SE = 12/sqrt(36) = 2. Z = (104-100)/2 = 2. For a two-sided standard normal test, p is about 0.0455. At alpha = 0.05, reject H0, assuming the test assumptions hold. This is not proof that every individual value differs from 100.

8.4 Chi-square test

The chi-square test of independence examines whether two categorical variables are associated in a contingency table. Expected count for a cell = (row total x column total) / grand total. Statistic: chi-square = sum((Observed - Expected)^2 / Expected). Degrees of freedom = (rows - 1)(columns - 1).

PYTHON / PSEUDOCODE

```
import pandas as pd
from scipy.stats import chi2_contingency
# rows = device; columns = conversion outcome
observed = [[40, 60], [30, 70]]
chi2, p, dof, expected = chi2_contingency(observed)
print(chi2, p, dof)
print(expected)
```

- Use counts, not percentages, as the contingency-table input.
- Observations should be independent; expected counts should be adequate for the approximation. Small expected counts may require an exact or alternative method.
- Rejecting independence indicates evidence of association, not that one variable caused the other.
- For goodness-of-fit, the question is whether observed category counts fit a specified distribution; this differs from independence testing.

Memory trick

Z-test: standardized difference. Chi-square independence: categorical x categorical, observed versus expected counts.

09. Predictive modelling and supervised segmentation

Predictive modelling learns patterns from historical examples to estimate an unknown/future outcome. The target definition and validation design are as important as the algorithm.

9.1 Model-building flow

- Define the prediction moment, target, prediction horizon, user, and cost of mistakes.
- Prepare data and split it before fitting preprocessing or selecting features.
- Build a simple baseline; select informative attributes using domain knowledge and training-fold evidence.
- Train candidate models; validate with a metric aligned to the use case.
- Inspect errors, segment behaviour, estimate probabilities if required, and test stability.
- Communicate limitations, deploy under approval, and monitor performance and drift.

9.2 Predictive modelling: benefits and challenges

Benefits	Challenges / limitations
Ranks risk, forecasts demand, detects likely churn/fraud, supports resource planning.	Historical patterns may not persist; labels may be biased; data can be missing or noisy.
Makes decisions more consistent and scalable.	Model can leak future information, overfit, or fail on unseen populations.
Can reveal useful patterns and support what-if decisions.	Prediction is not explanation or causation; governance, privacy, fairness, and monitoring matter.

9.3 Informative attributes and progressive segmentation

An informative attribute separates target outcomes in a useful way. For example, delivery delay may vary by region, then by courier within region. Progressive segmentation repeatedly partitions data using features that improve separation, but segments need enough observations and should be validated rather than built on noise.

9.4 Trees as a set of rules

PYTHON / PSEUDOCODE

```
if customer_tenure < 3 months:
    predict = "high churn risk"
elif support_tickets >= 4:
    predict = "high churn risk"
else:
    predict = "lower churn risk"
```

A decision tree applies a sequence of splits. Each root-to-leaf path can be read as an IF-THEN rule. Trees are interpretable but can be unstable and may overfit unless complexity is controlled.

9.5 Supervised segmentation

Supervised segmentation uses a known target label to form groups whose outcomes differ. This is not the same as unsupervised clustering, which groups observations without using a target label. A split should improve a criterion such as impurity reduction and be assessed on held-out data.

9.6 Probability estimation

A classifier may output a score or estimated probability, such as P(churn=1 | features). Check calibration if probability values inform decisions. A model can rank customers well yet be poorly calibrated.

9.7 Predictive vs prescriptive

Predictive	Prescriptive
Estimates outcome: which delivery may be late?	Chooses action: which shipments should receive extra capacity?
Output: score, class, probability, forecast.	Output: recommended action, schedule, allocation or policy.
Evaluated by predictive quality and generalization.	Evaluated by objective value, constraints, feasibility, and consequences.

10. Regression analysis and forecasting

Regression models a relationship between a target and one or more predictors. Forecasting predicts values indexed by time and must respect time order.

10.1 Simple linear regression

Model: y = beta0 + beta1*x + error. beta0 is the expected y when x=0 (if meaningful); beta1 is the expected change in y for a one-unit increase in x under the model. Fitted values are estimates, not guarantees for individual observations.

10.2 Regression workflow and assumptions

- Inspect relationship and scale; consider scatterplots and domain knowledge.
- Check linearity, residual pattern, independence, constant variance, influential points, and multicollinearity in multiple regression.
- Avoid extrapolating beyond the training range without a strong reason.
- Do not interpret association as causal effect unless the design and assumptions support causality.

10.3 Common regression metrics

Metric	Formula / behaviour	Best interpretation
MAE	Mean absolute error: mean(|y - yhat|)	Average absolute error in target units; robust compared with RMSE.
MSE	Mean squared error: mean((y-yhat)^2)	Penalizes large errors more; squared target units.
RMSE	sqrt(MSE)	Like MAE in target units; large errors receive extra penalty.
R-squared	1 - residual sum of squares / total sum of squares	Relative in-sample variance explanation vs mean baseline; can be negative on test data.

10.4 Forecasting

Method	Use / warning
Naive	Next value equals latest observation; useful baseline.
Seasonal naive	Forecast repeats value from same season in previous cycle.
Moving average	Smooths short-term noise; lags turning points.
Exponential smoothing	Weights recent observations more; variants handle level, trend, seasonality.
Regression with time/features	Can include calendar, promotion, price, or external variables; future feature values must be known/forecast.

Time-series leakage warning

For forecasting, split by time. Randomly shuffling time series can let future information enter training data and produce unrealistic scores.

PYTHON / PSEUDOCODE

```
# Conceptual chronological split
train = df[df["date"] < "2026-01-01"]
test  = df[df["date"] >= "2026-01-01"]
# Fit preprocessing/model on train only; assess on future period.
```

11. Simulation, risk analysis, and optimization

Simulation evaluates outcomes under uncertainty. Optimization searches for a decision that maximizes or minimizes an objective subject to constraints.

11.1 Simulation

A Monte Carlo simulation repeatedly samples uncertain inputs from chosen distributions, runs a model, and summarizes the resulting outcomes. It can estimate a range of outcomes, tail risk, and the probability of crossing a threshold. Quality depends on realistic input distributions and dependencies.

PYTHON / PSEUDOCODE

```
import numpy as np
rng = np.random.default_rng(42)
# illustrative demand and unit margin
n_sim = 10000
demand = rng.normal(loc=1000, scale=150, size=n_sim)
margin = rng.normal(loc=12, scale=1.5, size=n_sim)
profit = demand * margin
print(np.mean(profit), np.percentile(profit, [5, 50, 95]))
```

11.2 Risk analysis

- Identify uncertain inputs, their distributions, and dependencies.
- Define risk measures: probability of loss, downside quantile, expected shortfall, or threshold exceedance.
- Run scenarios and sensitivity analysis to identify the variables that drive outcomes.
- Report assumptions and uncertainty instead of presenting one simulated number as certain.

11.3 Linear optimization

A linear program optimizes a linear objective subject to linear constraints. Example: maximize profit = 5x + 4y subject to 2x + y <= 100, x + 2y <= 80, and x,y >= 0. Variables are decisions; objective represents what is optimized; constraints represent limited resources or rules.

11.4 Nonlinear optimization

Nonlinear optimization has a nonlinear objective, nonlinear constraints, or both. A local method may find a local optimum rather than the global optimum. Feasibility, starting values, scaling, and constraint handling matter.

Term	Meaning
Decision variable	Quantity the decision-maker can control.
Objective function	Mathematical representation of the desired outcome.
Constraint	Limit, capacity, policy, or requirement.
Feasible solution	Satisfies all constraints.
Optimal solution	Best objective value among feasible solutions under the model.
Sensitivity analysis	How results change when inputs, assumptions, or constraints change.

Link the methods

Predictive model: estimate demand. Simulation: explore uncertainty around demand and profit. Optimization: choose inventory or staffing under capacity constraints.

12. Overfitting, generalization, holdout, and cross-validation

The goal is not to memorize the training set; it is to perform well on data that the model has not seen and that resembles the future use case.

12.1 Underfitting vs overfitting

Pattern	Likely meaning	Possible response
Poor train and validation performance	Underfitting, weak features, noisy target, or poor setup.	Try better features/model, reduce excessive regularization, or revisit data/target.
Excellent train, poor validation	Overfitting, leakage, or train-validation distribution mismatch.	Simplify/regularize, add data, improve split, remove leakage, tune with CV.
Good train and validation, poor later production	Drift, changing population/process, or evaluation mismatch.	Monitor inputs/outcomes; retrain or redesign when justified.

12.2 Holdout vs cross-validation

Holdout	K-fold cross-validation
One train/validation split; fast and easy to explain.	Split into k folds; train/validate repeatedly and average scores.
Result can vary greatly with a small dataset or unlucky split.	More stable estimate, though more computationally expensive.
Keep a final test set untouched until final evaluation.	Hyperparameter tuning on CV means CV score is not an unbiased final report; preserve a test set if possible.

12.3 Leakage prevention

- Split before fitting imputation, scaling, feature selection, target encoding, or other learned preprocessing.
- Use a Pipeline so preprocessing is fitted inside each training fold.
- Do not use fields unavailable at prediction time (e.g. a post-outcome status).
- For groups (patient, user, device), use group-aware splits where required.
- For time-dependent prediction, use chronological splits and prevent future features from entering the past.

PYTHON / PSEUDOCODE

```
from sklearn.model_selection import cross_val_score, StratifiedKFold
from sklearn.pipeline import make_pipeline
from sklearn.preprocessing import StandardScaler
from sklearn.linear_model import LogisticRegression
pipe = make_pipeline(StandardScaler(), LogisticRegression(max_iter=1000))
cv = StratifiedKFold(n_splits=5, shuffle=True, random_state=42)
scores = cross_val_score(pipe, X, y, cv=cv, scoring="roc_auc")
print(scores.mean(), scores.std())
```

Exam rule

Fit preprocessing on training data only. Pipeline + correct split is the safer default for cross-validation.

13. Decision analytics and evaluating classifiers

Classifier evaluation starts with the decision and the cost of errors. Accuracy alone can be misleading, particularly with imbalanced classes.

13.1 Confusion matrix

	Predicted positive	Predicted negative
Actually positive	TP: true positive	FN: false negative
Actually negative	FP: false positive	TN: true negative

13.2 Metrics

Metric	Formula	Interpretation
Accuracy	(TP+TN) / total	Overall fraction correct; can hide minority-class failure.
Precision	TP / (TP+FP)	When model predicts positive, how often is it correct?
Recall / sensitivity	TP / (TP+FN)	How many actual positives are found?
Specificity	TN / (TN+FP)	How many actual negatives are found?
F1	2 x precision x recall / (precision+recall)	Harmonic mean of precision and recall.
ROC-AUC	Area under TPR vs FPR curve	Ranking separation across thresholds; may look strong under imbalance.
PR-AUC	Area under precision-recall curve	Often informative for rare positive classes.

13.3 Threshold and costs

The threshold converts a probability/score to a class. Lowering the threshold generally increases recall and false positives; raising it generally increases precision while missing more positives. Choose a threshold using validation data and business costs, not the test set.

13.4 Baseline and analytical framework

- Define objective and what a useful decision means.
- Choose a simple baseline: majority class, naive forecast, or simple business rule.
- Use a validation design that matches deployment, then report multiple metrics and uncertainty.
- Translate metrics into consequences: false alarms, missed cases, capacity, money, customer impact.
- Check segment-level performance, calibration, stability, fairness, and operational feasibility.

Worked imbalanced example

Suppose 1,000 transactions include 20 fraud cases. A model predicts every transaction as non-fraud and reaches 98% accuracy, but fraud recall is 0%. This demonstrates why the baseline and class-specific metrics matter.

Investment and performance implications

Model quality has value only if it improves a decision enough to justify development, data, infrastructure, review, and error costs. Estimate value under realistic conditions; avoid equating a small metric improvement automatically with business value.

14. Evidence, probability combination, and probabilistic reasoning

Probabilistic reasoning makes uncertainty explicit. Evidence updates belief only to the extent that the evidence is informative under the assumed model.

14.1 Bayes update as a reasoning template

- Prior: what was believed before observing the new evidence?
- Likelihood: how probable is this evidence under each possible hypothesis?
- Evidence: how probable is the observation across the competing hypotheses?
- Posterior: revised probability after using the evidence.

Posterior odds = prior odds x likelihood ratio. This form is useful when evidence can be compared through likelihood ratios. If evidence items are conditionally independent given a hypothesis, likelihood ratios may be multiplied; dependence between evidence items must not be ignored.

14.2 Evidence combination cautions

- Do not double-count evidence derived from the same original source.
- Check base rates and whether the likelihoods are calibrated for the current population.
- Distinguish P(evidence | hypothesis) from P(hypothesis | evidence).
- A posterior depends on assumptions and prior information; show sensitivity where decisions are high stakes.

Mini-example

A spam word can increase the probability that an email is spam, but the word is not proof. Its effect depends on how common spam is and how often the word appears in spam vs legitimate email.

15. Factor analysis, directional data, and functional data

The supplied syllabus names these three areas but gives no subtopic list. The notes below are introductory orientation, not a claim that every detail is prescribed for the exam.

15.1 Factor analysis

Factor analysis models correlations among observed variables using a smaller number of latent (unobserved) factors plus variable-specific error. For example, survey questions about satisfaction, trust, and repeat intent may load on one or more underlying constructs.

- Purpose: discover/estimate latent dimensions behind correlated measures and reduce complexity.
- Factor loading: strength/direction of association between an observed variable and a factor.
- Communality: share of a variable’s variance explained by common factors; uniqueness is variable-specific variance.
- Factor analysis is not the same as PCA: PCA constructs components to explain total variance; common factor analysis models shared variance.
- Interpretation depends on measurement quality, factor retention choices, rotation, and sample adequacy.

15.2 Directional data analytics

Directional data are values whose meaning depends on direction or angle, such as wind direction, compass bearings, or orientations. Because angles wrap around, ordinary arithmetic can be misleading: the mean of 359 degrees and 1 degree should be near 0 degrees, not 180 degrees.

PYTHON / PSEUDOCODE

```
import numpy as np
angles_deg = np.array([359, 1])
angles = np.deg2rad(angles_deg)
mean_angle = np.arctan2(np.sin(angles).mean(), np.cos(angles).mean())
print(np.rad2deg(mean_angle) % 360)  # near 0 degrees
```

15.3 Functional data analysis

Functional data analysis treats each observation as a curve or function, for example a patient's daily glucose curve, a machine's sensor trace, or a temperature profile over a day. It considers the whole trajectory, including shape and change over a domain, rather than treating every time point as unrelated columns.

- Typical concerns: align curves, smooth noisy observations, compare shapes, identify functional principal components, and predict trajectories.
- Ordinary tabular methods may ignore smoothness and dependence between nearby time points.
- Choose methods appropriate to sampling frequency, missing intervals, alignment, and the scientific question.

16. Power BI: architecture, Power Query, reports, and dashboards

Power BI turns data preparation and analysis into interactive reports. A reliable report starts with a clear question, a trustworthy data model, and useful visual choices.

16.1 Typical workflow

- Connect to source data: files, databases, or supported services.
- Use Power Query to import, clean, filter, reshape, merge, and append data (ETL).
- Build relationships and a semantic data model with appropriate data types and measures.
- Create visuals and report pages; add slicers/filters, titles, and context.
- Publish/share through the appropriate Power BI environment and permissions.
- Refresh data and validate that reports remain correct.

16.2 Common terms

Term	Meaning
Power BI Desktop	Authoring environment for data connection, model, DAX, and reports.
Power Query	Data preparation and transformation layer; steps are recorded for refresh.
Data model	Tables, relationships, measures, and semantic structure.
DAX	Expression language for measures and calculated columns/tables.
Report	One or more interactive pages of visuals.
Dashboard	Service-based canvas of pinned tiles; conceptually distinct from a multi-page report.
Refresh	Reload/update data according to configured source and permissions.

16.3 Visual design checks

- Each page should answer a small number of business questions.
- Choose a chart that matches the comparison; make units and time ranges explicit.
- Use consistent colors, sensible scales, and readable labels; avoid decorative clutter.
- Show filters and definitions clearly; verify totals reconcile to source data.
- Consider access permissions, data sensitivity, refresh failures, and performance.

17. Python lab patterns and reusable workflow

The syllabus labs emphasize Pandas, NumPy, scikit-learn, Matplotlib/Seaborn, SciPy, probability, descriptive measures, covariance/correlation, outliers, and Power BI.

17.1 Load and inspect

PYTHON / PSEUDOCODE

```
import pandas as pd
import numpy as np
import matplotlib.pyplot as plt
import seaborn as sns

df = pd.read_csv("data.csv")
print(df.shape)
print(df.head())
print(df.info())
print(df.isna().sum())
print(df.describe(include="all"))
```

17.2 Descriptive statistics and visualization

PYTHON / PSEUDOCODE

```
col = "amount"
print(df[col].mean(), df[col].median(), df[col].std())
print(df[col].quantile([0.25, 0.50, 0.75]))

sns.histplot(data=df, x=col, kde=True)
plt.title("Distribution of amount")
plt.show()

sns.boxplot(data=df, x=col)
plt.show()
```

17.3 Covariance and correlation

PYTHON / PSEUDOCODE

```
numeric = df.select_dtypes(include="number")
print(numeric.cov())
print(numeric.corr(method="pearson"))
sns.heatmap(numeric.corr(), annot=True, cmap="coolwarm", center=0)
plt.show()
```

17.4 Missing values and duplicates

PYTHON / PSEUDOCODE

```
# Decide treatment using context, not a blind default
print(df.isna().mean().sort_values(ascending=False))
print("duplicate rows:", df.duplicated().sum())
# Example only: remove fully duplicated records if justified
# df = df.drop_duplicates()
# Example only: numeric median imputation if suitable
# df["amount"] = df["amount"].fillna(df["amount"].median())
```

17.5 IQR outlier flags

PYTHON / PSEUDOCODE

```
q1 = df["amount"].quantile(0.25)
q3 = df["amount"].quantile(0.75)
iqr = q3 - q1
mask = (df["amount"] < q1 - 1.5*iqr) | (df["amount"] > q3 + 1.5*iqr)
print(df.loc[mask, "amount"])
# Flag and investigate before removing or capping.
```

17.6 Split, fit, evaluate - safer model pattern

PYTHON / PSEUDOCODE

```
from sklearn.model_selection import train_test_split
from sklearn.pipeline import make_pipeline
from sklearn.impute import SimpleImputer
from sklearn.preprocessing import StandardScaler
from sklearn.linear_model import LogisticRegression
from sklearn.metrics import classification_report

X = df.drop(columns=["target"])
y = df["target"]
X_train, X_test, y_train, y_test = train_test_split(
    X, y, test_size=0.2, random_state=42, stratify=y)
# This exact pipeline is for numeric features only.
model = make_pipeline(SimpleImputer(strategy="median"),
                      StandardScaler(),
                      LogisticRegression(max_iter=1000))
model.fit(X_train, y_train)
pred = model.predict(X_test)
print(classification_report(y_test, pred))
```

Lab warning

The example pipeline assumes numeric predictors. Categorical columns require an appropriate encoding strategy; mixed-type datasets usually need a ColumnTransformer. Never include the target column among the predictors.

17.7 General-purpose interpretation template

“The analysis used [population/sample] from [source]. After checking missingness, duplicates, and ranges, I explored [variables] with [statistics/plots]. The result suggests [pattern], measured by [statistic/metric]. The result is limited by [assumptions, sample design, possible confounders]. The next step is [test/validation/business action].”

18. Exam formula sheet and common traps

Memorize the meaning and assumptions together with each formula. A correct formula applied to the wrong question is still a wrong answer.

Topic	Formula / fact to recall
Mean	x-bar = sum(xi)/n
Sample variance	s^2 = sum((xi - x-bar)^2)/(n-1)
Standard deviation	s = sqrt(s^2)
IQR and fences	IQR = Q3-Q1; lower = Q1-1.5IQR; upper = Q3+1.5IQR
Coefficient of variation	CV = SD/mean x 100% (meaningful for suitable positive ratio-scale data)
Conditional probability	P(A|B) = P(A and B)/P(B)
Bayes theorem	P(A|B) = P(B|A)P(A)/P(B)
Binomial mean / variance	np; np(1-p)
Poisson mean / variance	lambda; lambda
Z-score	z=(x-mu)/sigma
SE of mean	sigma/sqrt(n), or s/sqrt(n) estimate
Pearson correlation	r in [-1,+1]; summarizes linear association
Z test for mean (known sigma)	Z=(x-bar-mu0)/(sigma/sqrt(n))
Chi-square independence	sum((O-E)^2/E); df=(r-1)(c-1)
MAE / RMSE	MAE=mean(abs(error)); RMSE=sqrt(mean(error^2))
Precision / recall	Precision=TP/(TP+FP); Recall=TP/(TP+FN)

Topic	Formula / fact to recall
F1	2PR/(P+R)
Accuracy	(TP+TN)/total

Frequent exam traps

- Fail to reject H0 is not the same as proving H0.
- A statistically significant result can have a tiny, irrelevant effect.
- P(A|B) is not P(B|A).
- Correlation is not causation; a third variable can explain an association.
- High accuracy can be useless on an imbalanced dataset.
- An IQR-flagged observation is a candidate to investigate, not automatically an error.
- CLT describes a sampling distribution (often of means), not necessarily the raw data.
- Cross-validation does not repair leakage if preprocessing was fit before the split.
- For forecasting, validate on future periods, not randomly shuffled data.
- Predictive modelling estimates what may happen; prescriptive modelling chooses an action under objectives/constraints.

19. Interview-style questions with compact answers

Use these to practise explaining concepts in two or three sentences, then add an example.

- Q1. What is the difference between analysis and reporting?

Answer: Reporting summarizes what happened in a repeatable format; analysis explores why it happened, what may happen, or what action should be taken.

- Q2. Why use median instead of mean?

Answer: Median is resistant to extreme values and often represents a skewed distribution better; mean uses all numerical values and is useful for symmetric data.

- Q3. What is the difference between population and sample variance?

Answer: Population variance divides squared deviations by N; the usual unbiased sample variance estimate divides by n-1 (Bessel correction).

- Q4. What does p-value mean?

Answer: Assuming the null model and assumptions, it is the probability of a result at least as extreme as observed; it is not the probability that H0 is true.

- Q5. What is the difference between covariance and correlation?

Answer: Covariance is unit-dependent; correlation standardizes association to a range from -1 to +1 and describes linear association when Pearson r is used.

- Q6. What is the central limit theorem?

Answer: Under suitable conditions, the sampling distribution of the sample mean becomes approximately normal as n increases; it does not require the raw data themselves to become normal.

- Q7. Why can accuracy mislead?

Answer: If the positive class is rare, predicting the majority class can yield high accuracy but zero recall on the class of interest.

- Q8. How do you avoid data leakage?

Answer: Split appropriately first; fit preprocessing and feature selection only within training folds; exclude fields unavailable at prediction time.

- Q9. What is overfitting?

Answer: A model fits training-specific noise and performs poorly on new data; compare training and validation results and control complexity.

- Q10. Holdout vs cross-validation?

Answer: Holdout uses one split; k-fold CV repeats training/validation across k folds, often giving a more stable estimate at higher cost.

- Q11. Predictive vs prescriptive analytics?

Answer: Predictive estimates an outcome; prescriptive recommends an action based on objectives, constraints, and consequences.

- Q12. What does Pearson r = 0 mean?

Answer: It indicates little or no linear association in the sample; nonlinear association may still be present.

- Q13. What is Simpson’s paradox?

Answer: An aggregate association can reverse after stratifying by a confounding/grouping variable, so examine relevant subgroups.

- Q14. When is a chi-square independence test used?

Answer: To assess association between two categorical variables using observed and expected counts in a contingency table.

- Q15. What is a baseline?

Answer: A simple reference method, such as majority class or naive forecast, against which model value is evaluated.

- Q16. What is calibration?

Answer: Among predictions assigned a given probability, the observed event frequency should be close to that probability.

- Q17. Why use a pipeline?

Answer: It keeps learned preprocessing inside each training split/fold and reduces leakage and reproducibility errors.

- Q18. What is a Monte Carlo simulation?

Answer: Repeatedly sample uncertain inputs, compute outputs, and study the resulting distribution and risk.

- Q19. Linear vs nonlinear optimization?

Answer: A linear program has a linear objective and linear constraints; nonlinear optimization includes nonlinear objectives or constraints.

- Q20. What does Power Query do?

Answer: It connects to sources and records data import/cleaning/transformation steps for use in the Power BI model.

20. Self-test: 30 exam-style MCQs

Try the questions without consulting the formula sheet. The answer key with short explanations follows the questions.

- 1. Which analytics type answers “what should we do next?”

A. Descriptive
B. Diagnostic
C. Predictive
D. Prescriptive

- 2. Which measure is generally least affected by an extreme high value?

A. Mean
B. Median
C. Range
D. Variance

- 3. IQR is equal to:

A. Q3 + Q1
B. Q3 - Q1
C. max - min
D. SD / mean

- 4. The usual sample variance denominator is:

A. n
B. n+1
C. n-1
D. sqrt(n)

- 5. P(A|B) is:

A. P(A and B)/P(B)
B. P(A)/P(B)
C. P(B|A)
D. always P(A)

- 6. A rare disease plus a test with 95% sensitivity does NOT imply every positive is 95% likely diseased because of:

A. Overfitting
B. Base-rate effect
C. IQR
D. CV

- 7. Number of successes in fixed independent trials with fixed p is often modelled by:

A. Normal
B. Binomial

C. Poisson only
D. Uniform

- 8. Under a Poisson model, mean and variance are both:

A. p
B. n
C. lambda
D. sigma

- 9. The CLT mostly concerns:

A. Raw data always normal
B. Sampling distribution of a statistic such as mean
C. Causal proof
D. Median only

- 10. Standard error of sample mean (sigma known) is:

A. sigma*n
B. sigma/sqrt(n)
C. sigma^2/n
D. n/sigma

- 11. Pearson correlation range is:

A. 0 to 1
B. -100 to 100
C. -1 to +1
D. unbounded

- 12. r near 0 guarantees no relationship:

A. True
B. False

- 13. A point outside IQR fences should be:

A. Always deleted
B. Investigated in context
C. Always winsorized
D. Automatically a typo

- 14. A p-value is calculated under the assumption that:

A. H1 is true
B. H0 and test assumptions hold
C. The effect is important
D. Sample has no noise

- 15. If p = 0.03 and alpha = 0.05, conventional decision is:

A. Accept H0 as true
B. Reject H0
C. Prove H1
D. Collect no further context ever

- 16. Chi-square independence is generally used for:

A. Two categorical variables
B. Two continuous measurements only
C. Forecasting time series
D. Linear regression residuals only

- 17. In decision tree rules, a root-to-leaf path can be expressed as:

A. An IF-THEN rule
B. A confidence interval only
C. A histogram
D. A covariance matrix

- 18. Predictive modelling primarily estimates:

A. Best action under constraints
B. Likely outcomes
C. Only past summaries

D. The true cause automatically

- 19. Prescriptive modelling adds:

A. Decision recommendations/objectives/constraints
B. Only data cleaning
C. No assumptions
D. A bar chart only

- 20. Which is suitable as a naive forecasting baseline?

A. Future value equals last observed value
B. Randomly shuffle time
C. Fit on test data
D. Use future labels as features

- 21. RMSE differs from MAE because RMSE:

A. Never uses errors
B. Penalizes large errors more via squaring
C. Is always lower
D. Has no units

- 22. Randomly shuffling time-series records for future forecasting evaluation can cause:

A. Better causality
B. Time leakage
C. Guaranteed calibration
D. Higher sample variance only

- 23. Excellent train performance but weak validation performance suggests:

A. Underfitting only
B. Overfitting or split mismatch
C. Perfect generalization
D. No need to evaluate

- 24. In cross-validation, scaling should be:

A. Fit on entire dataset before folds
B. Fit within each training fold via pipeline
C. Fit on test data
D. Skipped always

- 25. With 20 actual positive cases and none found, recall is:

A. 100%
B. 98%
C. 0%
D. 20%

- 26. Precision answers:

A. Among predicted positives, fraction actually positive
B. Among actual positives, fraction found
C. Overall accuracy
D. Fraction of negatives found

- 27. A classifier score of 0.9 is a calibrated probability only when:

A. It is the highest score
B. Model scores are calibrated and validation supports that interpretation
C. Accuracy > 90%
D. Target is numeric

- 28. Monte Carlo simulation is useful for:

A. Exploring outcomes under uncertain inputs
B. Proving an assumption true
C. Replacing all data validation
D. Removing all risk

- 29. Power Query in Power BI mainly supports:

A. Data import and transformations
B. Only writing Python classes

C. Hardware installation

D. Hypothesis acceptance

- 30. For angles 359 degrees and 1 degree, a circular mean is near:

A. 180 degrees

B. 360 degrees only

C. 0 degrees

D. 90 degrees

Answer key with reasons

- 1 D. Prescriptive analytics recommends actions.
- 2 B. Median is robust to a few extreme values.
- 3 B. IQR is the distance from Q1 to Q3.
- 4 C. n-1 is the usual Bessel-corrected denominator.
- 5 A. Conditional probability is joint probability divided by P(B).
- 6 B. The prior prevalence affects positive predictive value.
- 7 B. Binomial models successes in fixed independent trials.
- 8 C. Poisson mean and variance both equal lambda.
- 9 B. CLT concerns the sampling distribution, not raw-data normality.
- 10 B. Independent observations yield SE = sigma/sqrt(n).
- 11 C. Pearson r lies between -1 and +1.
- 12 B. Nonlinear patterns can have near-zero Pearson correlation.
- 13 B. An IQR flag triggers investigation, not automatic removal.
- 14 B. The p-value is evaluated under H0 and assumptions.
- 15 B. p < alpha leads to rejection of H0 under the chosen procedure.
- 16 A. A contingency table supports association testing for categories.
- 17 A. A path through tree conditions can be translated into an IF-THEN rule.
- 18 B. Predictive models estimate outcomes.
- 19 A. Prescriptive methods choose actions given objectives and constraints.
- 20 A. The last value is a valid naive benchmark for many series.
- 21 B. Squaring magnifies large errors before averaging.
- 22 B. Future information can contaminate the evaluation.
- 23 B. The train-validation gap often signals overfit or distribution mismatch.
- 24 B. Fold-specific fitting prevents information crossing validation boundaries.
- 25 C. Recall = TP/(TP+FN); TP = 0, so recall is zero.
- 26 A. Precision conditions on predicted positives.
- 27 B. A score is a probability only if its probabilistic interpretation is supported/calibrated.
- 28 A. Repeated random draws reveal a simulated outcome distribution.
- 29 A. Power Query imports, cleans, and reshapes data.
- 30 C. Angles wrap around; the circular mean is near 0 degrees.

21. Final checklist before the exam

Use this as the last pass. Be able to explain the reason behind each item, not just recite the name.

- Can I explain the analytics lifecycle in order and give one deliverable at each stage?
- Can I identify numeric, categorical, ordinal, binary, time, and directional data?
- Can I choose a useful plot for distribution, comparison, relationship, time trend, or categorical association?
- Can I calculate and interpret mean, median, quartiles, IQR, sample variance, SD, CV, and percentiles?
- Can I distinguish joint, marginal, and conditional probability and solve Bayes problems with base rates?
- Can I recognize Binomial, Poisson, and Normal settings and explain the CLT correctly?
- Can I describe sampling bias, standard error, confidence intervals, and resampling?
- Can I interpret covariance, Pearson correlation, skewness, outliers, and Simpson’s paradox?
- Can I write H0/H1, interpret p-values, distinguish Type I/II errors, and select Z or chi-square for suitable questions?
- Can I explain predictive versus prescriptive modelling, supervised segmentation, trees, and probability estimation?
- Can I calculate MAE/RMSE, use a forecasting baseline, and validate time series chronologically?
- Can I identify overfitting, prevent leakage, and explain holdout vs cross-validation?
- Can I compute confusion-matrix metrics and explain why accuracy can mislead?
- Can I describe Monte Carlo simulation, linear/nonlinear optimization, and sensitivity analysis?
- Can I explain the basic purpose of factor analysis, directional data analysis, functional data analysis, and Power Query?
- Can I reproduce the Pandas/SciPy/sklearn snippets and explain each line in plain English?

Final answer-writing structure

Define the concept -> state the formula/assumption -> solve or show the method -> interpret the outcome -> mention one limitation or trap.

Source basis: ACTS, Pune, “Suggested Teaching Guidelines for Data Analytics - PGCP-AI August 2026” (4-page PDF supplied in this conversation). Examples and revision aids are explanatory additions organized around its listed topics.

Final one-page recap: the analytics mindset

Use this as the last five-minute revision. In any analytics problem, explain not only which method you choose, but why it fits the question and what its result does - and does not - support.

The complete analytics loop

Remember the sequence

Business decision -> define target and success measure -> understand and clean data -> explore patterns -> choose a statistical or modelling method -> validate assumptions and performance -> communicate uncertainty and limitations -> act -> monitor.

High-value rules to remember

- Describe data before modelling it; check missing values, units, duplicates, class balance, and unusual values.
- Mean and standard deviation are sensitive to extremes; median and IQR are more robust.
- P(A|B) is not generally equal to P(B|A); include the prior/base rate when interpreting test results.
- Correlation is association, not proof of causation; investigate confounders and subgroup reversals.
- A p-value is conditional on H0 and test assumptions. Failing to reject H0 does not prove H0 true.
- Fit preprocessing on training data only. Prevent leakage and use a split that matches deployment, especially for time series or grouped data.
- Evaluate classifiers with the confusion matrix and task-relevant metrics; accuracy alone may hide minority-class failure.
- Compare a model with a baseline and translate metrics into practical cost, benefit, and uncertainty.

A reliable exam-answer template

Step	What to write
1. Define	State the concept and the business/statistical question.
2. Choose	Name the formula/test/model and why it fits the data type and goal.
3. Check	State key assumptions, sample design, and leakage/validity risks.
4. Calculate	Show the formula, substitute the values, and give the result.
5. Interpret	Explain the result in context and mention one limitation or next step.

Final reminder

A strong Data Analytics answer connects the question, the method, the evidence, and the decision. A number without interpretation is rarely the whole answer.